

Uncertainty-aware haplotype phasing of genetic and epigenetic variation from long reads

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Abstract

Long reads link genetic variants and native methylation, but turning them into reliable haplotypes remains difficult in repetitive regions. Here we present LongPhase 2, which co-phases SNVs, indels, structural variants and allele-specific methylation in one graph, uses quality- and context-aware weighting, and scores every variant's phasing uncertainty. A graph neural network then unphases, rather than flips, predicted misphased variants. On HG002 nanopore data (10–60×), scored against a benchmark built from a telomere-to-telomere assembly, LongPhase 2 makes 2.8- to 4.8-fold fewer switch errors than WhatsHap and HapCUT2 and runs six- to eightfold faster at 60×; co-phasing lengthens haplotype blocks by 40%. Against the earlier mapping-based benchmark, the same outputs differ only 1.1- to 1.7-fold, suggesting that such truth sets mask differences between methods. On PacBio HiFi data, without retraining, LongPhase 2 makes 2.5- to 3.5-fold fewer switch errors than WhatsHap. It is a drop-in replacement in Clair3, ClairS, Severus and EPI2ME.

Introduction

Haplotype phasing, the assignment of heterozygous variants to their parental chromosomes, is essential for interpreting the diploid human genome. Whether two variants act in *cis* or in *trans* decides compound heterozygosity in recessive disease¹, and allele-specific methylation on a phased haplotype reveals its parent of origin². Long-read sequencing has made read-based phasing practical at clinical and population scale. Individual reads span tens to hundreds of kilobases, link distant heterozygous sites directly and carry native base-modification signals; a single library thus yields haplotype blocks orders of magnitude longer than short reads allow^{3,4}. Read-based phasing also needs nothing beyond the sample's own reads: no sequencing of the parents, no population reference panel of the kind statistical phasing depends on⁵, and none of the additional Hi-C or Strand-seq libraries otherwise used to phase whole chromosomes⁶.

Read-based phasers optimize agreement between reads and candidate haplotypes through weighted minimum error correction or probabilistic models, as in WhatsHap⁷, HapCUT2⁸ and Margin⁹. These methods centre on single-nucleotide variants (SNVs), although WhatsHap, by re-aligning each read to both alleles, also co-phases small and large indels with them¹⁰. We previously introduced LongPhase¹¹, which recasts phasing as a weighted allele graph, resolves

haplotypes at chromosome scale and co-phases SNVs with structural variants (SVs)^{12,13}. WhatsHap and LongPhase now sit beneath much of the long-read analysis stack. Clair3 phases with WhatsHap by default and with LongPhase as an option, and the somatic callers ClairS and ClairS-TO use LongPhase by default^{14–17}; the EPI2ME wf-human-variation workflow phases every nanopore genome with LongPhase¹⁸; and the SV callers Sniffles2 and Severus take as input the haplotagged alignments that either phaser writes^{12,19}. An error made by the phaser is therefore inherited, without any warning, by every analysis downstream.

Three challenges remain in read-based phasing. The first is integrating 5-methylcytosine (5mC), which long reads report alongside the sequence, into haplotype inference. Where the two haplotypes differ in methylation, as at imprinted loci and at sites of sequence-dependent allele-specific methylation, a read’s CpG states tie it to one haplotype even where it spans no heterozygous variant, which is where sequence-based blocks end^{20,21}. Several methylation-aware tools use that signal once a sequence phase exists. NanoMethPhase assigns methylation to haplotypes that WhatsHap has already resolved²⁰; MethPhaser and HapBridge learn haplotype-specific CpGs from the reads an SNV phaser has tagged, then join adjacent blocks or flag suspected switches^{21,22}; and LongHap, which co-phases SNVs, indels and SVs, adds methylation only to bridge the gaps its sequence phasing leaves²³. Because the informative CpGs are defined by the preceding sequence phase, its errors propagate into the methylation evidence, and methylation can revise a phase only after it is built. Allele-specific methylation is moreover confined to a small fraction of CpGs, varies between cells and tissues, and is read as a per-molecule probability rather than a base^{20,21}; such sparse, noisy evidence is best weighed together with the sequence alleles on the same reads.

The second challenge is accounting for errors that depend on genomic context. WhatsHap weights observations in its minimum-error-correction objective, and HapCUT2 incorporates base-quality information in its likelihood model^{7,8}. Base quality alone, however, cannot establish that a read originates from the locus to which it is aligned. Base-calling errors concentrate in homopolymers and tandem repeats, reads misplaced in segmental duplications create spurious linkages, and heterozygous calls in copy-number-altered regions can mislead the phasing graph^{24,25}. These context-dependent errors can create switches within haplotype blocks and confound compound-heterozygosity calls and haplotype-aware somatic detection^{16,19}.

A third challenge is identifying unreliable phase assignments. Statistical phasers attach posterior probabilities to phased genotypes⁵, and HapCUT2 uses posterior confidence to prune variants and split low-confidence blocks⁸. Such confidence estimates depend on the phasing model and may remain high when misplaced reads provide internally consistent evidence. Local sequence context and the surrounding evidence therefore provide additional information for assessing phase reliability. WhatsHap and Margin report no confidence, and LongPhase reports only simple read support, the number of a read’s alleles on each haplotype and a Phred-scaled tagging quality derived from them, not the reliability of each variant’s phase. Learning-based phasers, built on graph neural networks or deep reinforcement learning^{26,27}, do not close this gap. They learn, partly from simulated reads, to optimize the same fragment-cut or minimum-error-correction objective as heuristic phasers, phase SNVs only and report no confidence, and their authors note that errors from mis-alignment in repetitive regions remain unresolved²⁷.

These limitations have been underestimated, because phasers are benchmarked against Genome in a Bottle (GIAB) truth sets built largely by mapping reads to GRCh38^{28,29}, which can assess only the regions where mapping-based methods already work³⁰. The HG002 v4.2.1 release omits both sex chromosomes and 12% of the autosomes, mostly satellite arrays and segmental duplications^{29,30}, where mis-mapped reads and collapsed paralogues create false linkages^{24,25}. Its phase is also partial and partly circular: it was recorded only for the 74% of heterozygous variants on which trio-based phasing agreed with Strand-seq and HiFi phasing by WhatsHap²⁹, leaving the variants hardest for a read-based phaser the least likely to be scored. Extending benchmarks into difficult sequence has changed conclusions before. GIAB’s growth from 85% to 92% of GRCh38 exposed eight times more false negatives in a short-read call set²⁹, and the complete CHM13 reference removed tens of thousands of spurious variant calls per sample³¹. The complete diploid T2T-HG002 assembly now lifts this ceiling for phasing³⁰. It resolves both haplotypes of every chromosome, adds 701 Mb of autosomal sequence absent from v4.2.1 and phases every heterozygous site. Its projection onto GRCh38, the GIAB v5.0q benchmark³², extends assessment into tandem repeats and homopolymers, although satellite arrays and many segmental duplications remain excluded.

Here we present LongPhase 2, which addresses each of these limitations. It places 5mC sites in the same weighted phasing graph as SNVs, small indels and SVs, letting methylation inform the phase while it is being built. It weights every read observation by base quality and sequence context, giving less weight to homopolymers, tandem repeats and copy-number-altered regions. It scores the phasing uncertainty of every variant, and an optional graph neural network unphases the variants it predicts to be misphased, leaving their genotypes unchanged. We evaluate it against both GIAB v4.2.1 and the T2T-HG002-derived benchmark, and it runs as a drop-in replacement for LongPhase in existing pipelines.

Results

LongPhase 2 combines joint phasing with learned error detection

LongPhase 2 constructs a single phasing graph per chromosome from SNVs, small indels, SVs and allele-specific 5mC sites identified by the built-in `modcall` module (Fig. 1). Quality-dependent weighting and context-specific filters reduce the influence of unreliable observations, including those in homopolymers, tandem repeats and candidate copy-number-altered regions (Fig. 1a,b). After haplotype assignment, each phased variant receives a phasing-uncertainty score. A graph neural network (GNN) then evaluates variants in local windows centred on sites with high uncertainty (Fig. 1c). Variants predicted to be misphased are unphased while their genotypes are retained; blocks disconnected by this removal are split. We refer to this step as GNN correction. The method outputs phased VCFs for each variant class and haplotagged alignments (Fig. 1d).

We evaluated LongPhase 2 using HG002 nanopore R10.4.1 data down-sampled to 10–60× and PacBio HiFi data from the same individual at 10–50× (Methods). Nanopore results at 10–20× are means of ten down-sampling replicates. Results at 30–60×, all HiFi results and analyses of unphased variants use one down-sampling per coverage. We assessed phasing against two HG002 benchmarks: GIAB v4.2.1, integrated from mapping-based call sets, and GIAB



Figure 1: **Overview of LongPhase 2.** **a**, SNVs, small indels, SVs and 5mC sites carried by the same long reads enter one phasing problem; alleles are coloured by read haplotype. **b**, Weighted phasing graph. Read-supported edges link allele nodes, and each observation is weighted by base quality and sequence context; the inset shows an indel down-weighted inside a tandem repeat. **c**, Each variant receives a phasing-uncertainty score (Methods), and a graph neural network (GATv2 and Transformer layers; Supplementary Fig. 6) classifies the variants around the most uncertain ones. Variants predicted to be misphased are unphased (grey), not flipped; we call this step GNN correction. **d**, Output: phased haplotypes of all four variant classes, with the misphased variant left unphased (faded), and haplotagging reads.

v5.0q, in which every heterozygous variant is phased using the complete diploid telomere-to-telomere (T2T) assembly. The benchmarks share 2.50 Gb of autosomal sequence. Relative to v4.2.1, v5.0q covers more tandem-repeat and homopolymer sequence but less segmental-duplication sequence (Methods). Unless otherwise stated, results describe SNV phasing against v5.0q, and LongPhase 2 denotes the complete method, including GNN correction. We first compare LongPhase 2 with existing phasers, then assess its individual components. Per-run values and summary metrics are provided in Supplementary Data 1 and Supplementary Tables 7–12.

Comparison with existing phasers

LongPhase 2 achieves 2.8- to 4.8-fold lower SNV switch error rates than WhatsHap and HapCUT2. Using identical PEPPER-Margin-DeepVariant calls for SNV-only phasing, LongPhase 2 achieved switch error rates of 0.082% at 10 $\times$ and 0.023% at 60 $\times$. The corresponding rates were 0.233% and 0.081% for WhatsHap 2.8, and 0.266% and 0.104% for HapCUT2 1.3.4 (replicate s.d. at 10 $\times$ $\leq$ 0.006 percentage points). Across 10–60 $\times$, rates were 2.8–3.7-fold lower than those of WhatsHap and 3.3–4.8-fold lower than those of HapCUT2 (Fig. 2a,b and Supplementary Table 8)[Required validation, not yet run. (i) Genome-wide rates include the 17 chromosomes from which training windows were drawn, on the same down-sampling replicates. Report metrics with and without GNN correction on the held-out chr17, chr21 and chr22 (switch errors, assessed pairs, retained variants, N50; longphase compare -regions) as the primary validation, and state whether these chromosomes influenced the graph constants. (ii) Matched-retention comparison: withhold the same number of variants by the network, by an entropy threshold, by a support-count threshold, at random and by HapCUT2’s per-variant confidence, and plot switch errors against retained variants and N50, separating unphasing from block splitting. (iii) Paired uncertainty (replicate-wise differences) for the fold changes at 10–20 $\times$.]. Hamming distance ranked the tools in the same order (Fig. 2c). The block N50 of LongPhase 2 was slightly lower than that of the comparators at 10–20 $\times$ (0.84 versus 0.95 Mb for WhatsHap at 10 $\times$) and slightly higher at 30–60 $\times$ (2.86 versus 2.72 Mb at 60 $\times$; Fig. 2d). The lower switch error rates were accompanied by a modest decrease in phased fraction: WhatsHap phased 0.7–1.0 percentage points more benchmark heterozygous SNVs (Fig. 2f). Whole-genome phasing took 3 min at 10 $\times$ and 15 min at 60 $\times$, compared with 26 and 95 min for HapCUT2 and 26 and 115 min for WhatsHap [Runtimes have no exact source: they were read from the earlier Fig. 2e (pptx), which the regenerated Fig. 2 no longer contains, and supplementary.xlsx holds none. Supply the timing table (threads, hardware, wall-clock and peak memory per tool and coverage) as a Supplementary Table, or remove this sentence and the speed claim in the abstract. The repository README times phase of v2.0 at 24 threads as 39 s at 10 $\times$ and 180 s at 60 $\times$, about five times faster than the values quoted here].

We also compared three additional phasers at 10–20 $\times$ (Fig. 2e and Supplementary Fig. 10). Ralphi²⁷ and GCphase³³ had switch error rates 2.5–3.5-fold and 21–26-fold higher than those of LongPhase 2, respectively[Hamming distances near 40% are close to random phasing; confirm that GCphase was run with the parameters its authors recommend for nanopore data before reporting it]. Margin⁹ had the lowest rate at 10 $\times$ (0.069% versus 0.082% for LongPhase 2),

accompanied by a 3.2-fold lower block N50 (0.26 versus 0.84 Mb). At 12–20×, LongPhase 2 had lower switch error rates and 2.1–2.8-fold higher block N50 values than Margin.

Benchmark choice changes the measured separation between phasers. Against GIAB v4.2.1, LongPhase 2, WhatsHap and HapCUT2 produced 1,600–2,200 switch errors at 30–60×, with a 1.2–1.3-fold difference between the highest and lowest rates. Against v5.0q, the same phased VCFs showed a 4.3–4.8-fold difference (Fig. 3a,b and Supplementary Table 9). The change was tool-dependent. HapCUT2 had more switch errors against v5.0q at every coverage, and WhatsHap at every coverage below 60×; both had 1.6-fold more at 10×. By contrast, LongPhase 2 had fewer switch errors against v5.0q (1,518 versus 1,820 at 10×; 496 versus 1,676 at 60×). Hamming distance changed little between benchmarks (Fig. 3c,d). The contributions of benchmark-region coverage, variant representation and phase discordance remain unresolved because the error locations have not been compared directly [Score the regions and alleles common to both benchmarks separately from the benchmark-specific ones, and intersect switch-error locations across tools and benchmarks].

LongPhase 2 retains lower SNV switch error rates when co-phasing indels. Enabling indel phasing in WhatsHap increased block N50 from 2.72 to 4.20 Mb at 60×, but increased SNV switch error rates by 17–20% across coverages (Fig. 4a,b). With the same indels, LongPhase 2 had 3.0–3.7-fold lower SNV switch error rates, whereas WhatsHap phased 2.4–4.5 percentage points more benchmark heterozygous indels (Fig. 4c) and achieved 14–28% higher block N50 values. Adding SV and 5mC evidence to LongPhase 2 reduced this contiguity difference: its block N50 was 2–19% lower than that of WhatsHap with indels (4.10 versus 4.20 Mb at 60×), while its SNV switch error rates remained 3.0–3.7-fold lower (Fig. 4b,d and Supplementary Table 10). The provisional MethPhaser²¹ comparison yielded the same block N50 and switch-error counts as its input LongPhase 2 SNV blocks, whereas joint SNV and 5mC phasing increased block N50 by 8% (Supplementary Fig. 11). Verification that MethPhaser completed and that its output was scored is required before interpreting this comparison [The MethPhaser values in MethPhaserCompare.jsx are identical, to every printed digit and at all six coverages, to the SNV-only LongPhase 2 run without GNN correction given to MethPhaser as input (Supplementary Data 1, replicate 1). Confirm that the MethPhaser output VCF was scored and that MethPhaser ran to completion (it is expected to merge blocks); if it did not, re-run it before this comparison is reported].

LongPhase 2 achieves 2.5- to 3.5-fold lower switch error rates on PacBio HiFi data. On HG002 PacBio HiFi data, LongPhase 2 had 2.5–3.5-fold lower SNV switch error rates than WhatsHap without retraining the nanopore-derived network (Fig. 5a,b and Supplementary Table 11). Rates were 0.052% versus 0.128% at 10× and 0.042% versus 0.125% at 50×. LongPhase 2 also had lower Hamming distances at every coverage (Fig. 5c). Block N50 differed by less than 3% between the tools, while WhatsHap phased 0.4–0.7 percentage points more benchmark SNVs (Fig. 5d,e) [The HiFi call sets are nearly identical at every coverage (2,367,609–2,367,619 heterozygous SNVs from 10 to 50×; Supplementary Table 7), which suggests that variants were called once and the same calls phased at every down-sampled coverage, unlike the nanopore

runs. Confirm, and state the HiFi data source, caller and procedure in Methods; if one call set was used, the phased fraction at 10× (87%, against 78% on nanopore) is not comparable between platforms]. At 50×, the LongPhase 2 block N50 was 4.6-fold lower for HiFi than for nanopore data (0.55 versus 2.51 Mb), consistent with the shorter HiFi reads.

LongPhase 2 and WhatsHap leave distinct, partly overlapping sets of SNVs unphased. At 60×, LongPhase 2 left 210,683 heterozygous SNV calls unphased, compared with 142,313 for WhatsHap. Of these, 88,692 were unphased in both outputs, 121,991 in LongPhase 2 alone and 53,621 in WhatsHap alone (Fig. 6a). Each tool-specific set was enriched in intervals containing switch disagreements between the other tool and the benchmark. Enrichment increased from 4.3-fold at 10× to 9.6-fold at 60× for SNVs unphased by LongPhase 2 alone, and from 2.0-fold to 9.5-fold for those unphased by WhatsHap alone (Fig. 6b). These enrichments describe regional co-location and do not establish the correctness of individual phase assignments. The sets also differed in read depth (Fig. 6c and Supplementary Fig. 12). At 60×, SNVs unphased by WhatsHap alone had a median depth of 18×, whereas 15.6% of those unphased by LongPhase 2 alone reached the variant caller’s depth cap[the median depths and the zero-entropy share among WhatsHap-only sites (57.6%) are quoted from JHL’s commit message of 2026-09-27 and are not in any committed table; add the per-set summary to Supplementary Data 1]. This contrast indicates that the tools withhold phase under different read-depth conditions; capped depths alone do not identify the source of excess read coverage.

Contribution of the internal modules

GNN correction reduces switch error rates with a modest loss of phased SNVs and contiguity. Before GNN correction, LongPhase 2 had SNV switch error rates of 0.121% at 10× and 0.034% at 60×, already 1.9–2.5-fold lower than those of WhatsHap and 2.2–3.3-fold lower than those of HapCUT2 across coverages (Supplementary Fig. 13 and Supplementary Table 8). GNN correction reduced these rates by a further 31–33%, while decreasing the phased fraction by 0.1–0.3 percentage points. Block N50 decreased by 5–9% at 10–20× and 2–5% at 30–60×. On HiFi data, GNN correction reduced switch-error counts by 19–28%, with decreases of 0.1 percentage points in phased fraction and 1–2% in block N50 (Supplementary Fig. 13e,f).

Methylation increases contiguity at similar switch error rates, whereas indels increase both contiguity and error. We evaluated seven combinations of variant classes, each with and without GNN correction (Supplementary Figs. 14 and 15 and Supplementary Table 10). Without correction, adding 5mC increased block N50 at 60× by 8%, SVs by 1%, indels by 49% and all three by 75% (2.91 to 5.10 Mb). Only indels increased the SNV switch error rate, from 0.034% to 0.043% at 60× and from 0.121% to 0.155% at 10×[For configurations that phase indels, compare computes the Hamming distance over all common heterozygous variants, indels included (CompareProcess.cpp:579), whereas switch errors are SNV-only. Every Hamming value of an indel-containing configuration in this section, in Figs. 3c,d and 4d and in Supplementary Tables 9 and 10, therefore mixes SNV and indel phasing. Re-score these runs with SNV-only Hamming distance and report indel phasing accuracy separately]. GNN cor-

rection reduced switch error rates by 30–42% across configurations and reduced the additional error associated with indels by 41–67%. Block N50 decreased by 12–22% in indel-containing configurations as unphasing disconnected previously bridged blocks.

After GNN correction, SNV+5mC phasing had a switch error rate similar to that of SNV-only phasing (497 versus 496 switch errors at 60×), with an 8% higher block N50. Four-class phasing yielded a switch error rate of 0.027% and a 43% higher block N50 than SNV-only phasing (4.10 versus 2.86 Mb). LongPhase 2 phased 30.8–45.5% of benchmark heterozygous indels, more than 99% of 5mC sites called by `modcall` and 68–70% of heterozygous SVs called by `Sniffles2`[The SV and 5mC phased fractions were read from the earlier Fig. 4c,d (pptx) and have no exact source: supplementary.xlsx lists only the called SV and 5mC totals (Supplementary Table 7). Add the phased SV and 5mC counts to Supplementary Data 1 or drop these two clauses].

SNVs unphased by GNN correction are enriched outside regions with one-to-one assembly alignments. GNN correction unphased approximately 56,000–64,000 SNVs per genome, including 56,076 at 60×. At 10×, 81% of unphased SNVs were absent from the v5.0q benchmark VCF, 5% were classified as homozygous and 14% (8,673) as heterozygous (Supplementary Fig. 16a)[The classification counts benchmark VCF records outside the benchmark BED, whereas `compare` scores only inside it: `compare` puts the loss of phased benchmark SNVs at 60× at 2,055 (2,193,366 before and 2,191,311 after GNN correction, Supplementary Data 1), and the assembly analysis below finds only 549 unphased SNVs inside the benchmark BED, of which 413 are benchmark variants. The confirmed total at 60× is 56,076; update the benchmark-classification counts in Supplementary Fig. 16a from the same confirmed set. State the matching rule and SNV filter of each analysis]. Absence from this VCF does not by itself establish that a call is erroneous, because the benchmark depends on assembly alignment and variant representation.

We next examined the assembly alignments in an exploratory analysis. Of the 56,076 SNVs unphased at 60×, 11.8% lay in regions where both assembled haplotypes aligned one-to-one to GRCh38, compared with 92.6% of a random sample of SNVs phased before correction (odds ratio for lying outside these regions, 94; Supplementary Fig. 16b and Supplementary Table 12). Enrichment outside one-to-one alignments was observed on every autosome (Supplementary Fig. 16c). Sites outside these regions clustered in pericentromeric satellites, particularly at chr11:51–54 Mb, chr12:35–36 Mb and chr18:15–20 Mb. Among the 6,601 unphased SNVs within one-to-one alignments, 3,421 (52%) had no assembly variant record at the same coordinate[v5.0q is derived from the same assembly; this analysis extends coverage beyond the benchmark BED; it is not independent confirmation. The `dipcall` BED is the raw alignment and does not exclude the assembly errors that the benchmark curation removes. Rebuild this analysis before submission from one site-level table: autosomes only (chrX and chrY are currently included); a control drawn from SNVs that stayed phased, excluding the removed ones (the current background is sampled from all SNVs phased before GNN correction, `run_t2t_validation.sh`:75); and an allele-aware comparison with the assembly VCF that handles overlapping complex variants (the current test matches the start coordinate only, line 104). Derive every panel of Fig. 6 from that table]. This analysis extends beyond the benchmark regions but uses the same underlying assembly, and its background includes sites subsequently unphased by the GNN. It therefore does not provide

independent validation of phase or allele identity. Among unphased benchmark heterozygotes, 43% at 10× and 22% at 60× had a phasing entropy of zero, which can indicate unanimous votes or no incoming votes at a block start (Methods). Among benchmark-absent calls, 46–49% were clustered within 50 bp or had low genotype quality (Supplementary Fig. 17)[these prevalences were read from the earlier Fig. 6b,c (pptx), now Supplementary Fig. 13; replace with exact values and regenerate the figure].

Discussion

LongPhase 2 integrates genetic variation and allele-specific methylation in a shared phasing graph, then withholds phase assignments predicted to be unreliable. Joint use of these evidence classes extends earlier approaches that use methylation or SVs to connect haplotype blocks in regions with few heterozygous SNVs^{2,11,20,21,34}. Whereas LongHap uses methylation to bridge blocks after sequence phasing²³, LongPhase 2 incorporates methylation and sequence evidence during haplotype construction. The evidence classes made distinct contributions: methylation increased contiguity at similar SNV switch error rates, whereas indels accounted for most of the increase in both contiguity and error. The corresponding trade-off in WhatsHap suggests that this limitation is not specific to LongPhase 2, although shared difficulties in representing or modelling indels may contribute. GNN correction reduced the additional error associated with indels, and improvements in indel calling could further improve this balance^{15,35}.

Selective withholding complements evidence integration by reducing reliance on uncertain phase assignments. LongPhase 2 reports phasing entropy and weighted vote totals as measures of the underlying evidence. These differ from the model-based confidence estimates provided by HapCUT2⁸ and statistical phasers⁵: when votes are present, low entropy indicates their agreement, not necessarily strong evidence or a low probability of error. The GNN incorporates neighbouring graph and sequence features, allowing it to identify candidates that a high-entropy threshold alone would miss. However, its added value over entropy thresholds, support-count thresholds, random withholding or HapCUT2's confidence pruning remains to be established at matched variant retention. Its phase-error score is neither calibrated nor included in the output. Because withholding variants and splitting blocks change which variant pairs can be assessed, reductions in switch error rate must be interpreted jointly with phased fraction and contiguity. Margin's lower error rate but shorter blocks at 10× illustrates this trade-off. Most SNVs unphased by the GNN lay outside regions with one-to-one assembly alignments, where the present analysis cannot establish whether their phase was correct.

Benchmark choice substantially altered the measured separation between methods. At 30–60×, the highest and lowest switch error rates differed by 1.2–1.3-fold against GIAB v4.2.1 but by 4.3–4.8-fold against v5.0q, although more than 96% of each benchmark's autosomal regions were shared. The original v5.0q evaluation focused on variant calling rather than phasing³²; our results show that benchmark choice also changes the measured separation between phasing methods in this data set. The greater separation against v5.0q cannot yet be attributed to its additional tandem-repeat and homopolymer coverage, its complete phasing of heterozygous variants, or differences in representation and phase within shared regions. Nor does greater separation alone establish that all additional disagreements are phasing errors. Evaluation

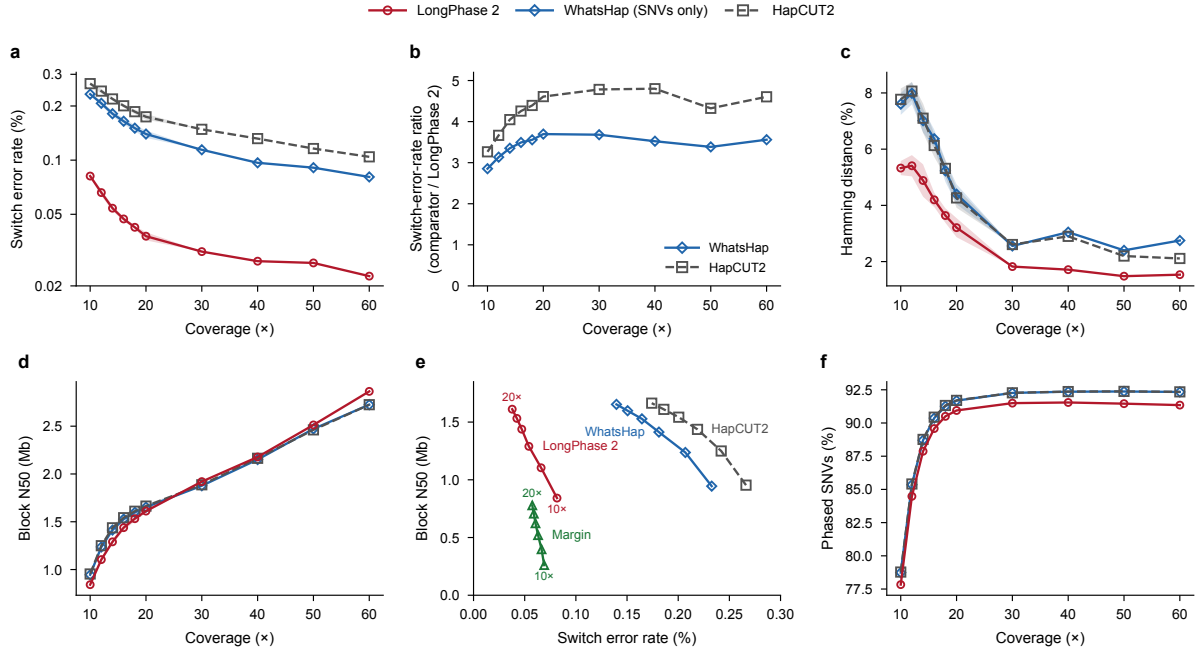


Figure 2: SNV phasing compared with WhatsHap, HapCUT2 and Margin. HG002 nanopore R10.4.1 reads down-sampled to 10–60×; SNV-only phasing of identical variant calls, scored against the T2T-HG002-derived v5.0q benchmark. Values at 10–20× are means of ten down-sampling replicates, with the s.d. as a light band (often narrower than the line); values at 30–60× come from one down-sampling of the 70× data set (Methods). Open symbols of different shape and size remain visible where tools coincide, as WhatsHap and HapCUT2 do in **d** and **f**. **a**, Switch error rate (log scale). **b**, Switch error rate of each comparator divided by that of LongPhase 2. **c**, Block-wise Hamming distance. **d**, Phase-block N50. **e**, Switch error rate against block N50 at 10–20× (means of ten replicates), one path per tool from 10× (bottom) to 20× (top), with Margin, which was run at these coverages only; Ralph and GCphase in Supplementary Fig. 10. **f**, Fraction of benchmark heterozygous SNVs phased.

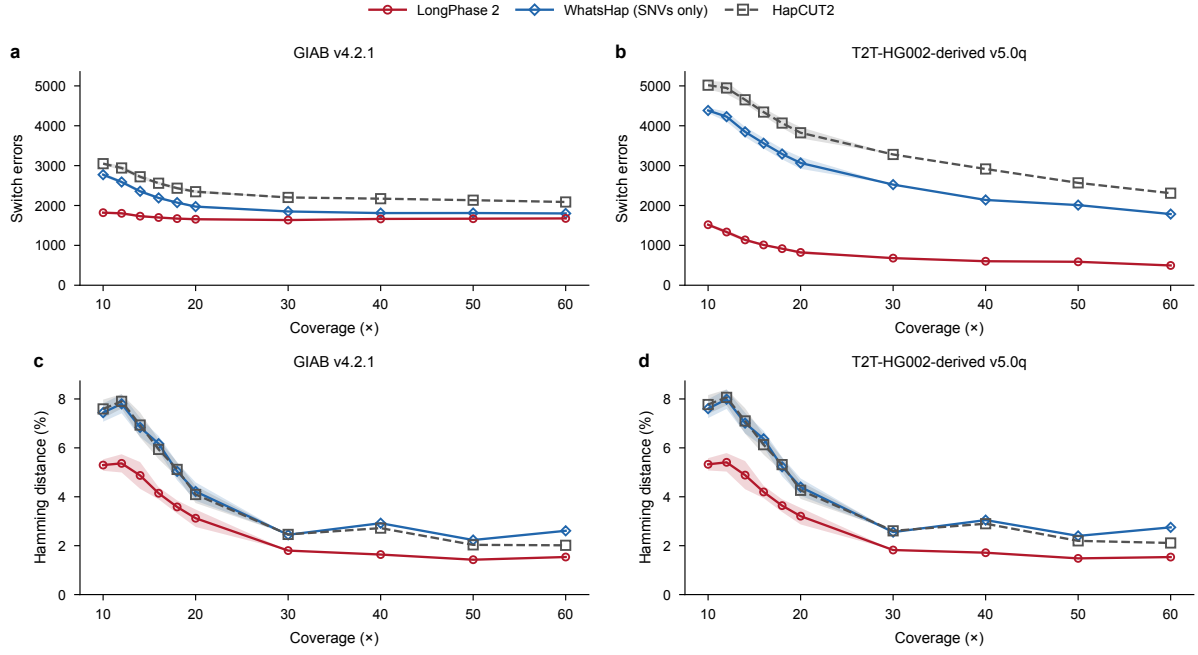


Figure 3: **The same phased VCFs scored against GIAB v4.2.1 and the T2T-HG002-derived v5.0q benchmark.** SNV-only phasing of HG002 nanopore R10.4.1 data; replicates as in Fig. 2. **a,b**, Genome-wide switch errors under v4.2.1 (**a**) and v5.0q (**b**), on a shared axis. **c,d**, Hamming distance under v4.2.1 (**c**) and v5.0q (**d**).

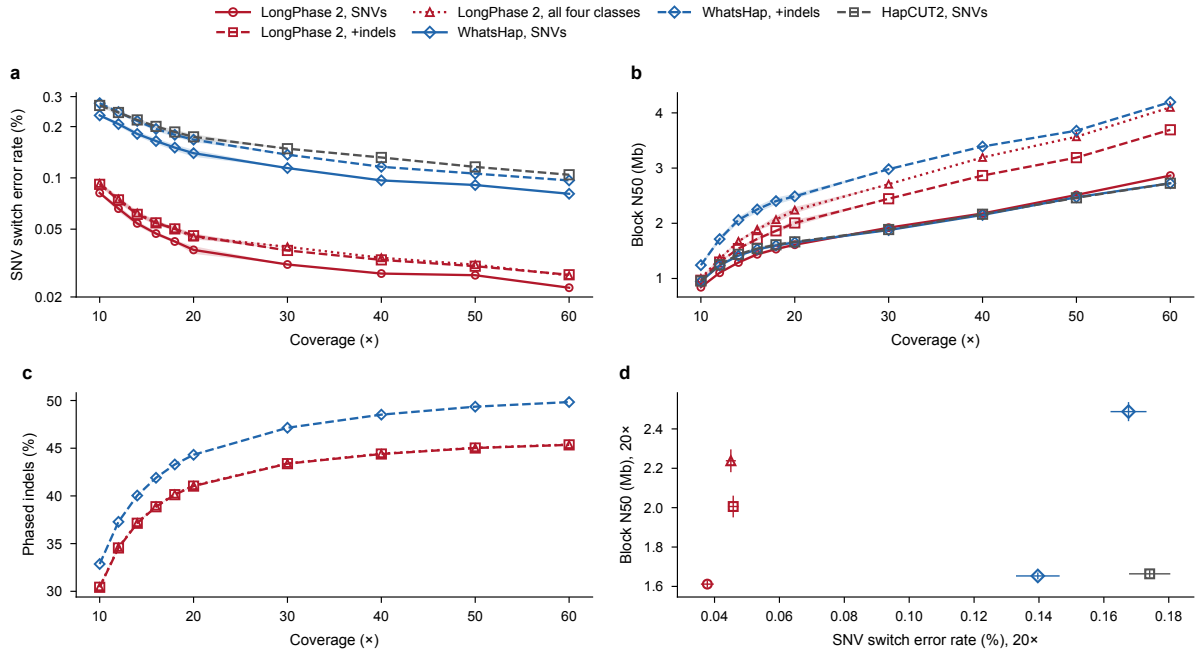


Figure 4: **Co-phasing compared with WhatsHap.** HG002 nanopore R10.4.1 data; LongPhase 2 phasing SNVs only, SNVs and indels, or all four classes (SNVs, indels, SVs and 5mC), against WhatsHap phasing SNVs only or SNVs and indels; HapCUT2 for reference. Accuracy is scored on SNVs against v5.0q; replicates as in Fig. 2. **a**, SNV switch error rate (log scale). **b**, Phase-block N50. **c**, Fraction of benchmark heterozygous indels phased. **d**, SNV switch error rate against block N50 at 20 $\times$ (means of ten replicates with s.d.).

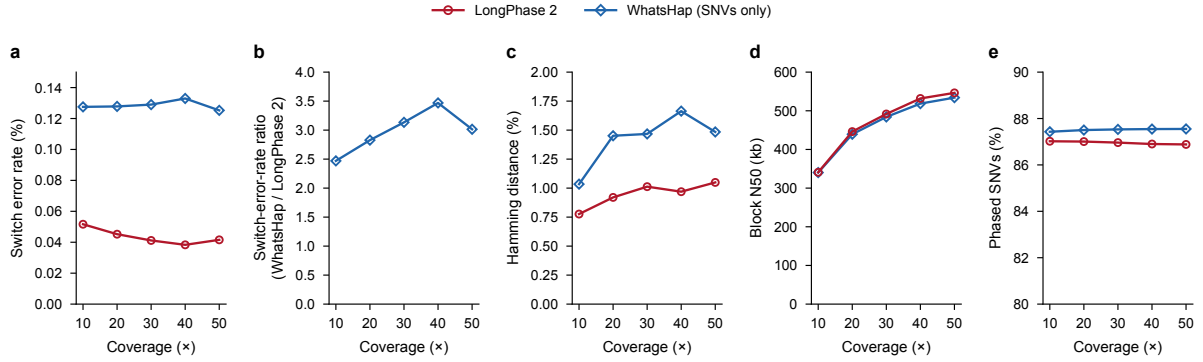


Figure 5: **Phasing of PacBio HiFi data compared with WhatsHap.** HG002 PacBio HiFi (Revio) reads down-sampled to 10–50 $\times$, one replicate per coverage[*state whether one call set was used for every coverage*]; SNV-only phasing scored against v5.0q; LongPhase 2 uses the network trained on nanopore data. Panels as in Fig. 2a–d,f. **a**, Switch error rate. **b**, Switch error rate of WhatsHap divided by that of LongPhase 2. **c**, Block-wise Hamming distance. **d**, Phase-block N50. **e**, Fraction of benchmark heterozygous SNVs phased.

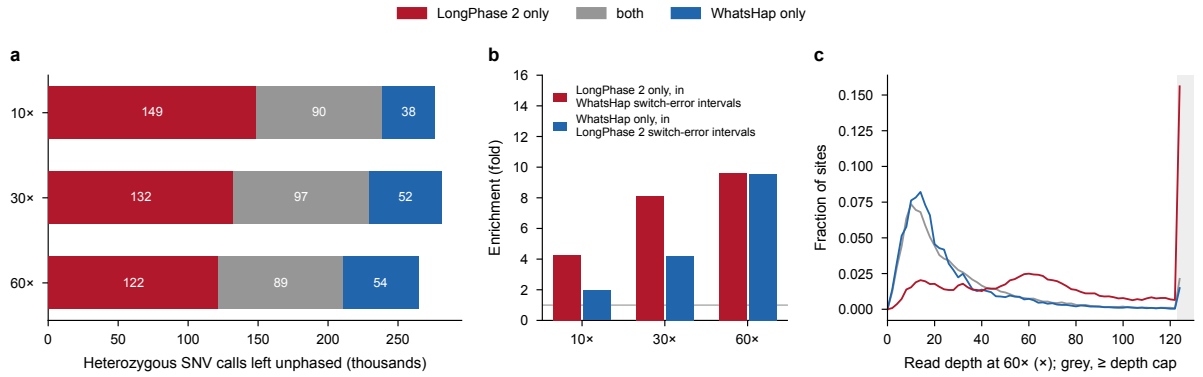


Figure 6: **SNVs left unphased by LongPhase 2 and by WhatsHap.** HG002 nanopore R10.4.1 data, SNV-only phasing, replicate 1; heterozygous SNV calls present in both output VCFs. **a**, Calls left unphased by LongPhase 2 only, by both tools or by WhatsHap only. **b**, Enrichment of each tool-specific set in genomic intervals containing switch disagreements between the other tool and the truth set, relative to that tool's genome-wide rate; a measure of regional co-location, not of site-level correctness. **c**, Read-depth distribution of the three sets at 60 $\times$ (2 $\times$ bins); the grey bin collects sites at or above the depth cap of the variant caller.

against both benchmarks, with separate analyses of shared and benchmark-specific regions and stratification by genomic context²⁵, is needed to distinguish these contributions.

The current evaluation has several limits. All data derive from HG002, and the network was trained on nanopore runs from the same down-sampling series, using labels from the evaluation benchmark. Genome-wide metrics include the 17 chromosomes that contributed training windows and therefore do not constitute an independent test of generalization. Phasing performance should be reported separately on held-out chromosomes and evaluated in other individuals against an independent benchmark, with model weights and thresholds fixed in advance. The HiFi results without retraining are consistent with transfer between sequencing platforms within HG002, but do not establish generalization to other individuals. Subsampling variability was not estimated for nanopore coverages above 20× or for HiFi, and HiFi call-set handling remains to be confirmed. For indels, SVs and 5mC, we measured phased fractions rather than class-specific phasing accuracy; SV and 5mC sites had no supervised training labels. Methylation co-phasing was assessed only on autosomes of a male lymphoblastoid cell line²¹, and the MethPhaser comparison remains provisional. The method also assumes a diploid genome with balanced allele fractions.

Compatibility with pipelines that use LongPhase^{14,16,18,19} provides a route to test the downstream value of selective phase withholding. In haplotype-aware somatic calling^{16,19}, reducing errors in germline phase could improve the assignment of tumour reads to haplotypes, but withholding phase may also reduce the information available to the caller. Varying the germline phaser while keeping the somatic caller fixed would quantify this trade-off. A calibrated per-variant confidence measure would further allow downstream methods to account explicitly for uncertainty in inherited phase.

Methods

Overview of LongPhase 2

LongPhase 2 is a diploid germline phaser that takes long-read alignments, a reference genome and heterozygous variant calls, and returns phased VCFs and haplotagged alignments (Fig. 1). SNVs, small indels, structural variants (SVs) and allele-specific 5-methylcytosine (5mC) observations are represented together in one weighted graph per chromosome. Initial haplotypes are resolved from weighted multi-neighbour votes, reads are used for a second phase-consistency pass, and a graph neural network (GNN) then scores local phase errors. Variants predicted to be unreliable are unphased, never flipped; if this disconnects a haplotype block, the surviving components receive updated phase-set identifiers. The detailed module rules are given in Supplementary Methods 1–6. The method extends the graph formulation of LongPhase 1¹¹.

Input variant calls

Heterozygous SNVs and small indels were taken from PEPPER-Margin-DeepVariant⁹; homozygous and filtered records were carried through unchanged but did not enter the graph. Indels in short tandem repeats were retained as evidence but down-weighted, because a misplaced repeat indel can link the wrong alleles. SVs were taken from Sniffles2¹² or cuteSV¹³ with supporting

read names; homozygous SVs, duplicated positions and SVs coincident with SNVs were excluded. The exact eligibility and repeat annotation rules are in Supplementary Method 2.

Calling allele-specific methylation

Modification probabilities were read from the MM/ML tags produced by Dorado³⁶ and carried through alignment. A CpG was called methylated when its 5mC probability was at least 0.8, unmethylated when it was at most 0.2, and ambiguous otherwise. Forward- and reverse-strand observations were mapped to one CpG coordinate. A site was retained as a candidate allele-specific site when methylated and unmethylated reads were sufficiently balanced and the ambiguous fraction was small:

$$\frac{\min(M_R, U_R)}{\max(M_R, U_R)} \geq 0.6, \quad \frac{A_R}{M_R + U_R + A_R} \leq 0.2, \quad (1)$$

where M_R , U_R and A_R are the three read counts. Candidates were then required to co-segregate with a neighbouring heterozygous marker. For a pair of sites, writing RR , RA , AR and AA for the four observed state combinations, linkage required

$$\frac{\max(RR + AA, RA + AR)}{RR + AA + RA + AR} \geq 0.9. \quad (2)$$

The same criterion was used for SNV-anchored and methylation-only linking. Only CpG dinucleotides were considered. Weak candidates were admitted in a later expansion pass, and consecutive CpGs were merged into one locus so that both strands shared one read list. The resulting methylated and unmethylated read names and depths were supplied to the phasing graph. The depth bound and expansion rules are in Supplementary Method 1.

Read-level allele observations and filtering

Alignments with mapping quality below 1, unmapped, secondary or duplicate records were excluded. SNV alleles came from the aligned base and its Phred quality; indel alleles from the CIGAR; SV alleles from a matching CIGAR event or the caller's read list; and 5mC alleles from the methylation read lists. Supplementary alignments were retained as evidence, so when two alignments of the same read overlapped by more than 20% of their combined span, the shorter alignment was discarded. Under the nanopore setting, a downstream SNV was removed when two SNVs at most 2 bp apart both lay in homopolymers of length at least three. A copy-number-aware filter identified intervals bounded by excess front- or back-clipping and removed variants whose alternate-supporting reads had an unusually high mismatch load relative to reference-supporting reads. Specifically, a read's mismatch load R_m counted the heterozygous sites in the interval at which it carried the alternate allele, and each variant was scored by

$$\text{MR}(v) = \frac{\bar{R}_m^{\text{ALT}}}{\bar{R}_m^{\text{REF}} + \bar{R}_m^{\text{ALT}}}, \quad (3)$$

where the barred terms are the mean loads of reads supporting the ALT and REF alleles of v . Sites with $\text{MR}(v) \geq 0.7$ were removed from all reads. The three-state interval scanner and its

constants are specified in Supplementary Method 2.

Weighted phasing graph

Each heterozygous variant contributed reference and alternate allele nodes. For each read, every allele was connected to the next 35 observed variants, with no edge-distance cap. Variants followed by a gap of more than 300 kb to the next heterozygous variant were skipped during haplotype resolution, casting no votes and receiving no haplotype or entropy. A read contributed weight 1 when both observations had quality at least 12 and weight 0.1 otherwise. Indel and 5mC observations and SV alternate observations were assigned fixed quality 60; reference-allele SV observations were assigned 30. Thus all such observations have full weight at the default threshold, while reference-SV evidence is down-weighted only at a higher threshold.

For a variant pair (u, v) , the four allele-pair supports were accumulated as w_{rr} , w_{ra} , w_{ar} and w_{aa} . The two possible pairings have support $P = w_{rr} + w_{aa}$ and $Q = w_{ra} + w_{ar}$, and their ambiguity is $s = \min(P, Q) / \max(P, Q)$.

Haplotype resolution by multi-neighbour voting

Variants were processed in coordinate order. Each variant cast a vote to each downstream neighbour for the pairing with larger support, except when $s > 0.7$; SNV–5mC pairs used a 0.3 threshold and required at least one read-equivalent of support. Votes had weight 1 by default, 20 for near-unanimous support ($s \leq 0.1$ with total support at least 1, or one pairing below 1 while the other was at least 1), and 0.1 when the voting variant was a tandem-repeat indel. The vote was translated using the already assigned haplotype of the upstream variant, producing totals $h_1(v)$ and $h_2(v)$. When more than three votes had support of at most 1 read-equivalent, and at least one qualifying vote remained, only votes with $s < 0.2$, weight at least 1 and a non-indel voting variant were retained. The larger total assigned the haplotype. A tie beyond the last accepted edge opened a new block; a tie within an existing block left the variant unphased and cast no votes. The first variant position defined the phase-set identifier. The constants were fixed by a pre-study sweep on HG002 at 10–60 $\times$ and were not re-tuned for the reported experiments; the calibration is described in Supplementary Method 3.

Read-based correction and phasing entropy

After the graph assignment, each read was assigned to a haplotype by a weighted majority of its phased SNV and SV alleles (weight 1), indel alleles (weight 0.1) and 5mC alleles (not used for this pass). Reads were retained only with a majority above 0.65 and total evidence above 1. For each variant, the consistency of the two read populations was

$$\rho = \frac{\max(n_1, n_2)}{n_1 + n_2},$$

where n_1 counts haplotype-1 reads carrying REF plus haplotype-2 reads carrying ALT, and n_2 counts the converse. The majority pairing assigned the phase only when $\rho > 0.75$; otherwise the variant was left unphased.

For each phased variant, phasing entropy was calculated from the weighted haplotype totals after the single-read guard:

$$\text{PE}(v) = - \sum_{i \in \{1,2\}} p_i \log_2 p_i, \quad p_i = \frac{h_i}{h_1 + h_2}. \quad (4)$$

It ranges from 0 for unanimous support to 1 bit for equal support. When $h_1 + h_2 = 0$, PE was set to 0 and the case was identified from the accompanying h_1, h_2 fields; this occurs only at the first variant of a block. PE, h_1 and h_2 were written to the VCF, and accepted voting edges with their accumulated weights formed the GNN input graph.

Graph neural network for phase-error detection

The GNN operated on merged SNV/indel, SV and 5mC records. A window of at most 41 variants (82 allele nodes) was opened around every phased variant with $\text{PE} \geq 0.8$, extending 20 variants on either side. It assigned a non-calibrated phase-error score to each variant. The 31 node features summarized phasing evidence, sequence context, graph structure and the four variant classes; six edge features summarized vote weight, genomic distance, allele matching and phase-set relations. Edges were restricted to the window, symmetrized and supplemented with mean-feature self-loops. Exact definitions and normalizations are in Supplementary Tables 1–2.

The network used four GPS graph-transformer layers³⁷: a local four-head GATv2 branch^{38,39} over graph edges and a global four-head self-attention branch⁴⁰, followed by residual normalization and a 128–256–128 feed-forward block. A two-layer classifier produced the two-class softmax output. The two allele-node scores were averaged, as were scores from overlapping windows. Only variants no further from the trigger than half the largest trigger distance in the window were scored; the same rule defined the nodes used for training loss. Nodes outside that zone supplied context only. Exhaustive architecture and inference details are in Supplementary Method 4 and Table 3.

Training windows came from the same HG002 nanopore co-phasing runs used for evaluation at 10–20× (six coverages and ten down-sampling replicates, 60 runs). Of 722,116 windows, chr1–14 and chr18–20 supplied training (602,788), chr15–16 validation (67,880), and chr17, chr21 and chr22 testing (51,448); the chromosome sets were disjoint. No test-chromosome window entered gradient updates, batch-normalization statistics or checkpoint selection. Windows contained all four variant classes, but only benchmarked SNV and indel nodes had labels, so SV and 5mC correction was transferred from neighbouring small-variant evidence. The training and evaluation truth was the v5.0q small-variant benchmark. Training windows were limited to 10–20×, whereas results also included 30–60× and PacBio HiFi data.

Labels described local phase orientation. Within each window and phase set, the first benchmarked variant anchored the orientation; later disagreements were labelled as a single-variant flip or unphase event, or as a break at the first site of a longer run. Benchmark-absent variants and SV/5mC records were masked from the loss but retained as graph context. The released model merged flip, unphase and break into one error class. Error labels accounted for 2.96% of labelled central nodes in the training split. Correct nodes were subsampled to at most four per error node, and the resulting cross-entropy used focal weighting⁴¹ with $\gamma = 1.5$ and la-

bel smoothing 0.05. Training used haplotype-swap and coverage-scaling augmentations; the optimizer, stopping rule and environment are listed in Supplementary Table 4.

Checkpoints were selected on chr15–16 by maximizing

$$S = R \left(2 - \frac{1}{P} \right) \quad (5)$$

over 81 score thresholds from 0.01 to 0.99, subject to $R \geq 0.005$, where P and R are error-class precision and recall. The selected checkpoint was epoch 44; validation and held-out classification metrics are given in Supplementary Table 4. The deployment threshold $B = 0.30$ was chosen empirically from a sweep of 0.05–0.95 across the 60 HG002 nanopore runs at 10–20 $\times$, balancing switch-error reduction against retention of phased SNVs. Compared with $B = 0.40$, the chosen threshold removed more switch errors at the cost of withholding additional phased SNVs, favouring greater error reduction while retaining most phased sites (Supplementary Method 4). [Provide the per-run threshold-sweep results and the phased-fraction/switch-error trade-off panel.]

GNN correction and phase-set update

A variant with mean score at least 0.30 was unphased while preserving its genotype. Bridge protection was disabled. If removing these variants disconnected a block, the surviving connected components became separate phase sets (Supplementary Method 4).

Haplotagging

Reads were tagged from the phased SNV, indel, SV and 5mC alleles they carried. Each informative observation contributed one vote, and a read was assigned to the majority haplotype when its fraction was at least 0.60; reads whose informative variants belonged to multiple phase sets or lacked a majority were left untagged. Tagged reads received HP, PS and phasing-quality fields. SV alleles were re-derived from the CIGAR and 5mC evidence was matched by read name. Full evidence and quality definitions are in Supplementary Method 5.

Benchmark data

The benchmark used HG002 (NA24385) Oxford Nanopore R10.4.1 reads from the ONT giab_lsk114_2022.12 release: two PromethION flow cells (PAG65784 and PAG68757), SQK-LSK114, 400 bases per second and 4-kHz sampling, basecalled with Dorado dna_r10.4.1_e8.2_400bps_sup@v4.0.0 and 5mCG_5hmCG@v2; their read N50 values were 29.2 and 29.4 kb. Reads were aligned to the GRCh38 no-alt analysis set with minimap2 v2.30. The 70.4 $\times$ alignment was down-sampled to 10, 12, 14, 16, 18, 20, 30, 40, 50 and 60 $\times$ with samtools v1.23.1⁴². Ten replicates were used at 10–20 $\times$ and one at 30–60 $\times$; variants were called separately at every coverage and replicate. SNVs and indels came from PEPPER-Margin-DeepVariant, SVs from Sniffles2 v2.8.0, and allele-specific 5mC sites from LongPhase 2. [Give the ten down-sampling seeds and the PEPPER-Margin-DeepVariant release/model, and state that any phased caller output was discarded.]

Accuracy was assessed against two HG002 small-variant benchmarks: GIAB v4.2.1²⁹, integrated from mapping-based call sets, and GIAB v5.0q³², derived from the diploid T2T-HG002

assembly³⁰, in which every heterozygous variant is phased. Each benchmark was evaluated within its own benchmark regions on autosomes 1–22. Of these regions, 2,496 Mb are common to both; v5.0q additionally covers 83 Mb, including 16 Mb of tandem repeats and 13 Mb of homopolymers, whereas the 46 Mb covered only by v4.2.1 include 29 Mb of segmental duplications. Satellite sequence is largely excluded from both (less than 3 Mb each), as determined with the GIAB v3.6 genome stratifications²⁵.

Comparison with other methods

WhatsHap v2.8⁷ and HapCUT2 v1.3.4⁸ were run on the same alignments and variant calls; cross-tool SNV comparisons used SNV-only phasing, with indel phasing disabled in WhatsHap. LongPhase 1.0¹¹ and LongPhase 2 were run with the nanopore setting, and GNN correction used PE trigger 0.8, score threshold 0.30, no bridge protection and block splitting. Co-phasing experiments separately evaluated the seven variant-class combinations listed in Supplementary Table 10, with and without correction. Indel co-phasing was also compared with WhatsHap’s default indel phasing. Margin v2.3.1⁹, Ralph²⁷, GCphase³³ and MethPhaser²¹ were evaluated as described in Supplementary Table 5. HG002 PacBio HiFi (Revio) reads were down-sampled to 10, 20, 30, 40 and 50× and phased SNV-only with LongPhase 2, with and without correction, and WhatsHap; the GNN weights were trained on nanopore data. [Give complete comparison commands and versions, state whether HapCUT2 default likelihood pruning was enabled, whether all tools used the same machine, and whether LongPhase 1.0 was re-run for every low-coverage replicate; document the HiFi source, alignment, caller, mode and call-set handling.]

Evaluation metrics

Metrics (Supplementary Fig. 8) were computed by intersecting heterozygous query and truth records by position and REF/ALT alleles, then partitioning the intersection into blocks phased in both files; singleton blocks were ignored. Within each block, switch errors were the Hamming distance between truth and query switch-encoded haplotype strings, and the switch-error rate divided their number by the assessed consecutive SNV pairs. Isolated switches and two-consecutive-switch flips were reported separately. Hamming distance was the block-wise minimum over the two orientations, divided by the number of compared variants; unlike switch rate, it included indels when present in both files. Phased fraction was the number of truth heterozygous SNVs or indels that were phased by the query divided by all truth heterozygous variants of that class. Block N50 was the span-weighted median of query phase-set spans of at least two variants, pooled over chromosomes. Switch-error positions were exported for the unphased-SNV analysis. The exact block and denominator definitions are in Supplementary Method 6. [State the exact region-filter commands and add a metric table defining truth eligibility, numerator, denominator and block handling; re-score indel-containing configurations with an SNV-only Hamming distance; define the depth, informativeness and genotype-quality bins used in Supplementary Fig. 17.]

Analysis of unphased SNVs

An SNV was counted as unphased when it was phased in the pre-correction output and unphased after GNN correction. Such sites were classified against v5.0q as absent, homozygous or heterozygous (Supplementary Fig. 16a). At 60×, they were also classified against the T2T-HG002 assembly projected to GRCh38 as outside the 1:1 assembly BED, inside without an assembly variant at the same start coordinate, or inside with one. A random sample of 50,000 SNVs phased before correction provided the background; enrichment was summarized by chromosome and 1-Mb bins. To compare tools, shared heterozygous SNVs were partitioned into sites unphased by LongPhase 2 only, WhatsHap only or both, and the fraction of each tool-specific set within the other tool's switch-error intervals was compared with the fraction of that other tool's assessed SNVs within those intervals. [Define switch-error intervals and deposit the scripts for Fig. 6 and Supplementary Fig. 16a.]

Runtime and memory

Wall-clock time and peak resident memory were measured with `/usr/bin/time -v`; reported LongPhase 2 times included alignment reading, graph construction, phasing, GNN correction and VCF writing, and excluded alignment, variant calling and down-sampling. [Record the machine, compiler, thread count, timing repeats and summary statistic, and report memory with the runtime results.]

Statistics and reproducibility

Ten nanopore down-sampling replicates at 10–20× were summarized by their mean and standard deviation, while 30–60× and HiFi used one down-sampling per coverage. Subsamples for each platform were drawn from the same source sequencing data set of HG002. The reported standard deviations therefore describe variation due to read subsampling, conditional on that data set, rather than variation among independent sequencing experiments. Subsampling variability was not estimated at 30–60× or for HiFi. Tools compared at each coverage and replicate used the same sampled reads. No statistical tests, sample-size calculation, randomization or blinding were used. GNN inference used fixed model weights and no sampling. [State the HiFi source depth and whether further replicates were possible, and report paired replicate-wise correction effects where applicable.]

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data analysed in this study are publicly available: the GRCh38 no-alt analysis set from NCBI (https://ftp.ncbi.nlm.nih.gov/genomes/all/GCA/000/001/405/GCA_000001405.15_GRCh38/seqs_for_alignment_pipelines.ucsc_ids/); the GIAB HG002 v4.2.1 (<https://ftp-trace.ncbi.nlm.nih.gov/ReferenceSamples/giab/release/AshkenazimTrio/>)

HG002_NA24385_son/NISTv4.2.1/GRCh38/) and v5.0q (https://ftp-trace.ncbi.nlm.nih.gov/ReferenceSamples/giab/release/AshkenazimTrio/HG002_NA24385_son/v5.0q/) benchmarks from the GIAB FTP site; and the HG002 nanopore R10.4.1 reads and alignments from the ONT open-data archive (s3://ont-open-data/giab_lsk114_2022.12/) under a CC BY-NC 4.0 licence. The benchmark files used are listed in Supplementary Table 5. [State the command used to derive the v5.0q chr1–chr22 subset and provide source data for Figs. 2–6 and every Supplementary figure with plotted values, including any access restrictions.]

Code availability

LongPhase 2 is open source under the GNU General Public License v3.0 at <https://github.com/twolinin/longphase>; the repository includes a demonstration data set and workflow. The reported results correspond to release v2.1 (commit 46ba47025b2704ab00cf54a35d3d4fead3eb148c). The training chain, released model, plotting code and validation scripts are not yet deposited with the public repository. [Deposit the training and figure-generation files under an OSI-approved licence, parameterize development-machine paths, and cite the deposited archive here.]

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